

A Look at Mobile OS Security: Android vs iOS

When it comes to mobile operating systems, which one is the most secure? This article attempts to assess the security levels of Android and iOS, and arrive at which one fares better.



The war between mobile operating systems (MOS) started with the advent of smartphones and other mobile devices, and shows no signs of ending anytime soon. We have had MOSs going head-to-head in multiple fields: usability, extensions and apps, support, community, even eye-candy and appearance! However, one persistent area of debate is related to mobile security.

Before I begin, allow me to clarify one rather minor, yet oft-repeated issue—the “*Wait for the next update, when (the name of problem appears here) will be fixed*” rhetoric, which has not been considered in this article.

When you think of it, both Android and iOS were created with security in mind. While Android has the ever-powerful Linux base, Apple’s iOS, too, is not a minnow when it comes to out-of-the-box security standards. Yes, Android’s open source standards ensure that the MOS remains regularly updated and patched, but even Apple has not created a billion-dollar empire by shipping vulnerable OSs. To quote Symantec regarding the security of both Android and iOS, “The ostensible goals of the creators of iOS and Android [were] to make the platforms inherently secure, rather than to force users to rely upon third-party security software.” That statement comes from a company that earns its livelihood by providing

users with ‘third-party security’ solutions for their devices. Naturally, this cannot be untrue!

Still, both Android and iOS have faced their due share of criticism when it comes to security. For instance, Android is said to offer little or no protection against data integrity attacks, while iOS has poor scores when it comes to implementing permission-based access control. Arguably, it is claimed that iOS has a way better encryption mechanism than Android. Yet, such encryption does not mean that iOS is better than Android at preventing resource abuse. In fact, Android performs much better, simply because it has the ability to run apps in isolation, thereby limiting the damaging ability of a malicious app.

However, considering the fact that both Android and iOS have been designed for the end user, they have to make a trade off—for the casual user, a security norm of varying degree is often abstracted in order to create a more usable OS.

Some number-crunching

Both Android and iOS are extremely popular (oops, isn’t that obvious?). Both the operating systems currently have a huge number of applications for their users. Here is a rather simple table to help you assess the state of each operating system: